THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE, REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

LONGIDO DISTRICT, FORM FOUR PRE – MOCK EXAMINATION SYNDICATE

032 CHEMISTRY

Time: 3 Hours **Thursday, 06th August 2020**

Instructions

1. This paper consists of sections A, B and C.
2. Answer **all** questions from sections A and B, and **only one** question from section C.
3. All writings must be in blue or black pen, except drawings that must be in pencil.
4. Non – programmable calculators may be used if necessary.
5. Write your **examination number** on every page of your answer sheet.
6. The following constants may be used

   Atomic masses:

   Na = 23, C = 12, O = 16, H = 1, K = 39, Cu = 63.5,
   Cl = 35.5, Ca = 40, S = 32, N = 14, Mg = 24.

   Avogadro's number = 6.02 x 10^{23}.

   GMV at s.t.p = 22.4 dm^3.

   1 faraday = 96500coulombs.
   1 litre = 1 dm^3 = 1000 cm^3.

SECTION A (15 Marks)

Answer all questions in this section

1. For each of the items (i) – (x), choose the correct answer from the given alternatives and write its letter beside the item number in the answer sheet provided.

(i) Namayani saw the "explosive" sign on a box. She made the following possible interpretation. Which interpretation is most correct?

A. The box contains fire wood B. The box contains sodium chloride
C. The box contains petrol D. The box contains gun powder

E. The box contains insecticides

(ii) The property of metal to be drawn into wires is called:

A. Malleability B. Ductility C. Conductivity D. Electronegativity E. Periodicity

(iii) Which of the following solutions is the most concentrated?

A. 50g of $CaCO_3$ in $100cm^3$ of water B. 60g of NaCl in $200cm^3$ of water

C. 120g of KNO_3 in $100cm^3$ of water D. 50g of NaOH in $200cm^3$ of water

D. 80g of $MgCl_2$ in $150cm^3$ of water

(iv) The following metals form basic oxides, except:

A. Copper B. Magnesium C. Lead D. Iron E. Calcium

(v) When carbon burns in a limited supply of oxygen it forms:

A. Carbon dioxide B. Carbon monoxide C. Hydrogen carbonate
D. Carbonate E. Coke

(vi) What will happen when zinc is placed in aqueous copper (ii) sulphate?

A. Zinc atoms are oxidized B. Copper atoms are oxidized

C. Zinc ions are oxidized D. Copper ions are oxidized

E. Zinc atoms are reduced

(vii) The occurrence of two or more compounds with the same molecular formula but different structure formula is known as:

A. Amphoterism B. Allotropy C. Isotopy D. Polymorphism E. Isomerism

(viii) The following acid – base reactions are suitable for titration, except:

A. Potassium hydroxide with nitric acid B. Ammonium hydroxide with sulphuric acid

C. Sodium hydroxide with ethanoic acid D. Ammonium hydroxide with oxalic acid

E. Sodium hydroxide with hydrochloric acid

(ix) Which among the following halogens is a strongest oxidizing agent?

A. Bromine B. Chlorine C. Astatine D. Fluorine E. Iodine

(x) A gas was passed into a test tube containing lime water.The lime water turned milky. This gas was most likely:

A. Nitrogen monoxide B. Sulphur dioxide C. Oxygen

D. Carbon monoxide E. Carbon dioxide

2. Match the items in **list A** with the responses in **list B** by writing the letter of the correct response beside the item number in the answer sheet provided.

List A	List B
(i) Green – yellow gas which rapidly bleaches damp litmus paper	A. Nitrogen
(ii) A solid carbon dioxide	B. Hydrogen chloride gas
(iii) Turns brown on exposure to air	C. Aluminium and iron
(iv) Their carbonates do not exist	D. Test for sulphates
(v) Forms a white dense fume with ammonia	E. Silver and lead
	F. Sulphur dioxide gas
	G. Chlorine
	H. Dry ice

SECTION B (70 Marks)

Answer all questions in this section

3. The table below shows three allotropes of carbon

Allotropes	Characteristics
A	It is soft and slippery in nature
B	It has no clear crystalline structure
C	No delocalized electron

(a) Name the allotropes represented by the letters A, B and C

(b) (i) How does an allotrope B formed?

(ii) Tabulate any two differences between allotropes A and C

4. (a) Account for the following observations.

(i) Oxides of potassium and sodium can not be prepared by indirect method.

(ii) Elements in group VIII of the periodic table are called noble gases.

(iii) A luminous flame produces a lot of soot and smoke.

(b) (i) Calculate the percentage composition by mass of water in a hydrated

Sodium carbonate, $Na_2CO_3.5H_2O$

(ii) State any two disadvantages of hard water.

5. (a) Define the term radical

(b) A compound contains 68.54% carbon, 8.63% hydrogen and 22.83% oxygen. Calculate:

(i) Empirical formula

(ii) Molecular formula if its relative molecular weight is 104.

6. (a) (i) State the modern periodic law

(ii) What is periodicity?

(iii) Define the term "electronic configuration"

(b) Suggest the best method of separating each of the following mixtures:

(i) Grains from husks

(ii) Oil from cotton seeds

(iii) Ethanol from water

(iv) Iodine from sand

7. Halima , a form four student performed an experiment on volumetric analysis.

She dissolved 2.12g of a certain carbonate (X_2CO_3) in water to make 250cm^3 of solution.

She used 25cm^3 of this solution to titrate against 0.25M of hydrochloric acid and obtain

the results as shown in the table **JJ.**

Table JJ

EXPERIMENT	PILOT	1	2	3
Final burette reading (cm^3)	16.30	16.00	32.00	48.30
Initial burette reading (cm^3)	0.00	0.00	16.00	32.00
Volume of acid used (cm^3)				

(a) (i) Complete the table and find the average volume of acid used.

(ii) Write a balanced chemical equation for the reaction.

(iii) Calculate the molarity of a metal carbonate

(b) (i) Find the relative molecular mass of a metal carbonate.

(ii) Calculate the atomic mass of a metal represented by a letter X.

(iii) Write the electronic configuration of the metal ion.

8. (a) Complete and balance the following chemical reactions

(i) MgO(S) + H_2O(l) $\longrightarrow$
(ii) $AgNO_3$(aq) + $MgCl_2$(aq) $\longrightarrow$
(iii) SO_2(g) + O_2(g) $\longrightarrow$
(iv) NH_3(g) + HCl (g) $\longrightarrow$

(b) What volume of oxygen gas will be produced at STP when 12.25g of potassium chlorate is thermally decomposed?

9. (a) Define the following terms

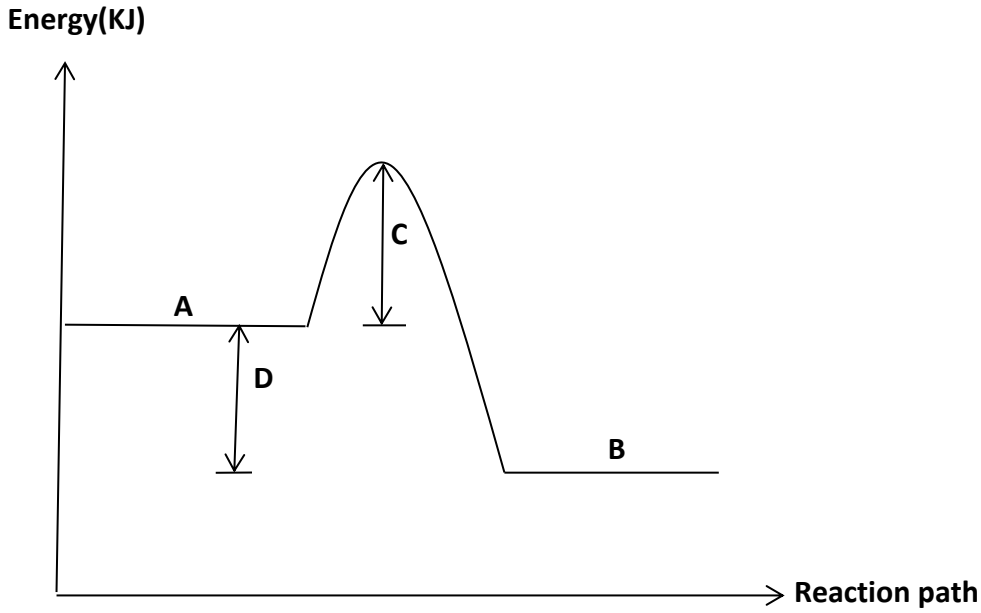
(i) Activation energy

(ii) Isolated system

(iii) Chemical kinetics

(b) State the Le chaterier's principle

(c) Nitrogen and hydrogen react to produce ammonia according to the following energy level diagram.



(i) What type of reaction is represented by this energy level diagram?

(ii) What do the letters A to D represent?

10. (a) State the faraday laws of electrolysis

(b) An electric current was passed in series through solutions of calcium chloride and copper (ii) sulphate. Carbon electrodes were used in both electrolytes.

If 1.4 litres of chlorine measured at STP were produced;

(i) What volume of oxygen at STP would also be produced?

(ii) What mass of copper would be produced?

11. (a) What is meant by the following terms?

(i) Fire fighting (ii) Striking back of a flame

(b) Briefly explain five effects of air pollution.

12. With the help of a well-drawn and labeled diagram, explain the process of extracting Sodium metal from its ore. Write all necessary equations for the reactions in each stage.

SECTION C (15 Marks)

Answer one question from this section

13. (a) Name the function group represented by each of the following general structures.

(i) R – COOH (ii) R – COOR (iii) R – OH

(b) Complete and balance the following organic reactions

(i) C_2H_2 + H_2O $\xrightarrow{H+}$

(ii) CH_3CH_2OH + KOH $\longrightarrow$

(iii) CH_3COOH + CH_3CH_2OH $\longrightarrow$

(c) By giving necessary conditions, reagents and balanced equations, explain how the following conversions take place.

(i) Methane to tetrachloro methane

(ii) Ethanol to ethane

(iii) Ethanol to ethanoic acid

14. (a) Define the following terms

(i) Soil profile (ii) Leaching (iv) Liming

(iv) Soil reaction (v) Fertilizer analysis

(b) Soil fertility does not necessary determine the soil productivity. Explain the correctness of this statement using four points.

(c) Assume you are a farmer and your farm has lost its fertility. Describe any six method that you can use to manage the loss of soil fertility.